LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

REG NO: 192324235 **NAME:** JOTHIKA M

EXP -10

10. Create a visual representation in Figma showcasing the differences between various multiple access techniques.

Aim: To create a visual representation in Figma showcasing the differences between various multiple access techniques.

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

Design:

Multiple Access Techniques in Mobile Communication

One radio medium, different ways to share it. Compare how users are separated by frequency, time, code, or a scheduled combination of subcarriers and time.

Overview ● | FDMA ↗ | TDMA ↗ | CDMA ↗ | OFDMA ↗

01 / FOUR WAYS TO SHARE THE AIR

A User A B User B C User C

FDMA

01 ↗

Frequency Division Multiple Access



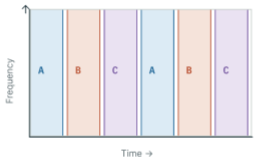
Separate frequency channels

Concurrent bands; gaps between bands are guard bands.

TDMA

02 ↗

Time Division Multiple Access



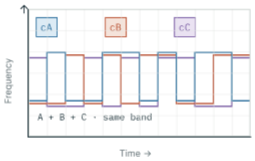
Successive time slots

A → B → C, then repeat; narrow gaps are slot guards.

CDMA

03 ↗

Code Division Multiple Access



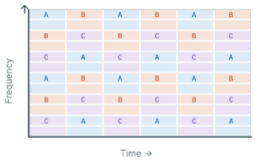
Concurrent coded signals

Codes overlap in the same time–frequency area, not exclusive tiles.

OFDMA

04 ↗

Orthogonal Frequency Division Multiple Access



Scheduled subcarrier groups

Narrow subcarriers are reassigned across time intervals.

In every diagram: frequency increases upward; time advances to the right. Color always identifies the same user, A, B or C. Labels provide a second identifier.

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Compare the engineering trade-offs

TABLE 01 / QUALITATIVE COMPARISON

Method	Resource separation	Synchronization needs	Efficiency considerations	Example technologies
FDMA	Frequency bands	Frequency accuracy and filtering; no shared time-slot alignment in basic FDMA.	Guard bands consume spectrum. Fixed allocations may sit idle when a user has no data.	Analog 1G AMPS.
TDMA	Time slots on a carrier	Tight burst timing and slot alignment; guard intervals absorb timing uncertainty.	Users share a carrier in bursts. Slot guards add overhead; users may wait for the next slot.	GSM: FDMA carrier channels combined with TDMA time slots.
CDMA	User-specific spreading codes	Code acquisition and tracking. User uplinks can be asynchronous; slot alignment is not universal.	Same-band reuse with interference averaging. Capacity is interference-limited and depends on power control and load.	IS-95 / cdmaOne; WCDMA / UMTS.
OFDMA	Orthogonal subcarriers × time	Tight time and frequency synchronization; sensitive to carrier-frequency offset.	Flexible, channel-aware scheduling. Cyclic prefix and pilots add overhead; high peak-to-average power ratio.	LTE downlink: OFDMA; uplink: SC-FDMA. 5G NR: CP-OFDM-based allocation; uplink also supports DFT-s-OFDM.

There is no universal efficiency ranking: allocation, traffic, channel conditions, overhead and implementation all matter. These methods are not interchangeable generational labels.

READING NOTE These are simplified conceptual resource-allocation views, not waveform spectra. Real systems combine techniques and differ by standard and link direction. Open each method note for its operating principle, benefits, limitations and use case.

FDMA

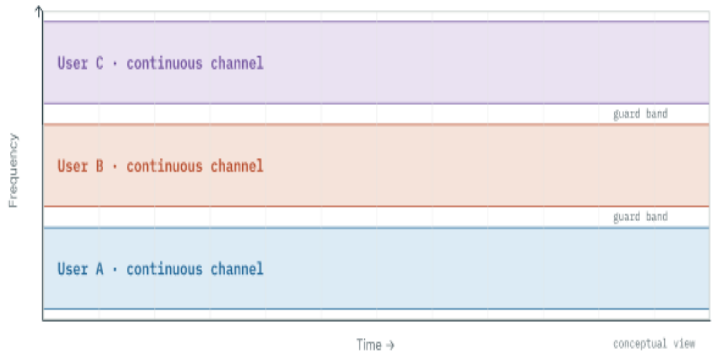
Frequency Division Multiple Access

Separate users by frequency; let each channel transmit continuously.

Overview ↗ | FDMA ● | TDMA ↗ | CDMA ↗ | OFDMA ↗

Different bands, the same time

FIG. 02



A User A B User B C User C

Each user retains a separate frequency band across the observation interval. The blank strips between channels are guard bands, not extra user channels.

Guard bands reduce adjacent-channel interference

BASIC PRINCIPLE

A channel of your own

Divide the available spectrum into separate frequency channels. A receiver selects the desired channel with frequency tuning and filtering.

HOW USERS SHARE

Parallel, not turn-taking

A, B and C transmit concurrently on different carrier frequencies. In a fixed allocation, an idle user's channel remains reserved rather than being automatically reused.

SYNCHRONIZATION

Keep the carriers accurate

Frequency accuracy and selective filtering are required. Basic FDMA does not require users to align transmissions to shared time slots.

Advantages

- Simple channel separation through tuning and filtering.
- Continuous transmission; no wait for a recurring time slot.

Limitations

- Guard bands reduce the bandwidth available for payload.
- Fixed allocations are inflexible; idle channels waste reserved resources.

Representative use case

- Analog 1G AMPS voice: individual calls use assigned frequency channels.
- AMPS also uses paired uplink/downlink bands (FDD), a separate duplexing choice.

READING NOTE Frequency separation is the multiple-access idea. It is distinct from duplexing, which separates uplink and downlink. The diagram is a simplified allocation view, not a plot of transmitted spectra.

TDMA — Time Division Multiple Access

METHOD 02 / RESOURCE SEPARATION

MOBILE COMMUNICATION

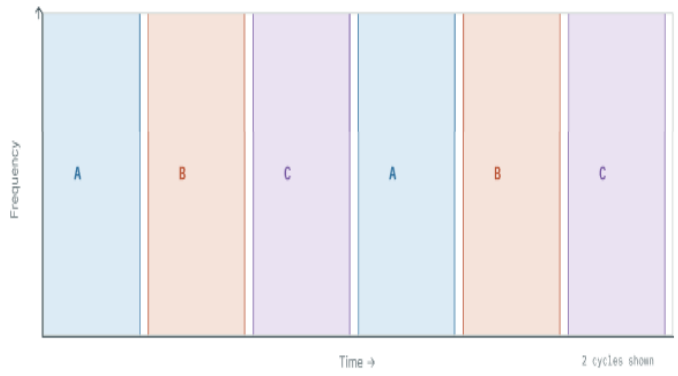
TDMA

Time Division Multiple Access

Share a carrier by assigning users successive, repeating time slots.

One carrier, different turns

FIG. 03



A User A B User B C User C

A → B → C repeats on one carrier. Each vertical block is a transmission burst; the narrow blank strips are slot guard intervals. Slot widths and guards are schematic.

Guard interval: time reserved between bursts

BASIC PRINCIPLE

Take turns on a carrier

Divide time into a recurring sequence of slots. A user transmits in its assigned slot rather than continuously occupying the shared carrier.

HOW USERS SHARE

Scheduled bursts

Users A, B and C use the same carrier at different times. The repeating schedule gives each user a predictable opportunity to send a burst.

SYNCHRONIZATION

Arrive inside the slot

Tight timing prevents bursts from overlapping at the receiver. Guard intervals and timing adjustment accommodate propagation and timing uncertainty.

Advantages

- Multiple users share one carrier through a clear schedule.
- Scheduled bursts allow predictable resource assignment.

Limitations

- Tight timing is needed; slot guard intervals add overhead.
- A user may have to wait for its next allocated slot.

Representative use case

- GSM combines FDMA carrier channels with TDMA slots.
- The figure shows the time-sharing idea on one carrier, not the entire GSM radio interface.

READING NOTE TDMA describes separation between users in time; it is not the same as TDD, which separates uplink and downlink in time. Systems may combine time sharing with frequency separation.

METHOD 03 / RESOURCE SEPARATION

MOBILE COMMUNICATION

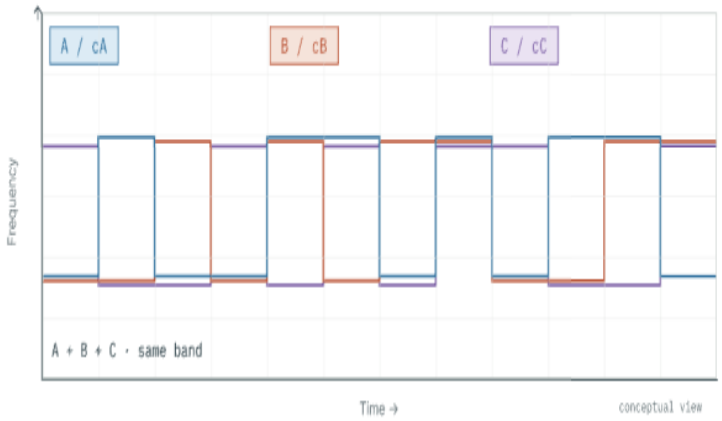
CDMA

Code Division Multiple Access

Share the same band at the same time; distinguish users by spreading code.

Same resources, different signatures

FIG. 04



A User A B User B C User C

All three code signatures occupy the same time-frequency area concurrently. The overlaid traces symbolize chip patterns; their vertical offsets do not assign separate frequency channels.

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A + B + C  → correlate with cA →  estimate of A
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BASIC PRINCIPLE

Identify the code

Spread each user's signal with a distinguishing sequence: cA, cB or cC. The receiver correlates with the desired code to despread that user's signal.

HOW USERS SHARE

Signals add in the channel

A, B and C transmit over the same band concurrently. Codes help the receiver separate the mixture; they do not eliminate all interference from other users.

SYNCHRONIZATION

Acquire and track the code

The receiver needs code acquisition and tracking. User uplinks can be asynchronous; a universal shared time-slot alignment is not a CDMA requirement.

Advantages

- Concurrent reuse of the same frequency band.
- Spreading provides interference averaging; performance still depends on load and channel conditions.

Limitations

- Near-far effects require careful power control.
- Multiple-access interference limits capacity; code processing adds receiver complexity.

Representative use case

- IS-95 / cdmaOne and WCDMA / UMTS use code-division multiple access.
- Their code structures and radio interfaces differ; the drawing is a shared conceptual model.

READING NOTE More codes do not mean unlimited users. Received power, code correlation, multipath, traffic load and receiver design determine how well simultaneous users can be separated.



Result: -

Hence created a visual representation in Figma showcasing the differences between various multiple access techniques.